Tutorial 5: Basic concept of chemical bonding

Part 1: Lewis symbol, Lewis structures, Resonance structures, exception of octet rule

Exercise 1. *(Und)*

a) Write Lewis structure that obeys the octet rule (duet rule for H) for each of the following molecules. Carbon is the central atom in CH_4, nitrogen is the central atom in NH_3, and oxygen is the central atom in H_2O.

a. F_2	e. NH_3
b. O_2	f. H_2O
c. CO	g. HF
d. CH_4	h. $SeCl_2$

b) Write the Lewis structures that obey the octet rule for the following species. Assign the oxidation numbers and formal charges for each atom. (i) OSC ; (ii) $SOCl_2$ (S is bonded to the two Cl atoms and to the O) ; (iii) BrO_3^- ; (iv) $HClO_2$ (H is bonded to O).

Exercise 2. *(Und)*

The most common type of exception to the octet rule are compounds or ions with central atoms having more than eight electrons around them. PF_5, SF_4, ClF_3 and Br_3^- are examples of this type of exception. Draw the Lewis structure for these compounds or ions. Which elements, when they have to, can have more than eight electrons around them? How is this rationalized?

Exercise 3 *(App)*

Consider the following bond lengths:

C—O 143 pm C═O 123 pm C≡O 109 pm

a. In the CO_3^{2-} ion, all three C—O bonds have identical bond lengths of 136 pm. Give the explanation?

b. Order the following species with respect to carbon-oxygen bond length (longest to shortest).

CO, CO_2, CO_3^{2-}, CH_3OH

What is the order from the weakest to the strongest carbon-oxygen bond? (CH_3OH exists as H_3C—OH)

Exercise 4. *(App)*

a) Use the concept of resonance to explain why all six C—C bonds in benzene are equal in length.

b) Are the C—C bond lengths in benzene shorter than C—C single bonds? Are they shorter than carbon-carbon double bonds?

Part 2: Ionic bond, covalent bond, Bond polarity

Exercise 5.

Determine whether each of the following compounds is likely to exist as a molecule, or as an ionic compound:

(i) HF ; (ii) $SiCl_4$; (iii) S_8 (elemental sulfur) ; (iv) Na_2O ; (v) PF_3 ; (vi) Be_3N_2

Exercise 6.

a) Which of the following bonds are polar:

(i) B—F (ii) Cl—Cl (iii) Se—O (iv) H—I

Which is the more electronegative atom in each polar bond?

b) The iodine monobromide molecule, IBr, has a bond length of 2.49 Å and a dipole moment of 1.21 D. Which atom of the molecule is expected to have a negative charge?

c) Calculate the effective charges on the I and Br atoms in IBr in units of the electronic charge, *e*.

Part 3: Bond energies

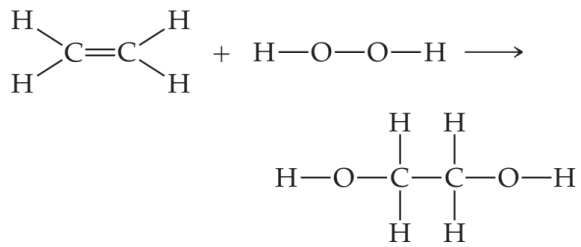
Table 8.4 Average Bond Enthalpies (kJ/mol)

Single Bonds							
C—H	413	N—H	391	O—H	463	F—F	155
C—C	348	N—N	163	O—O	146		
C—N	293	N—O	201	O—F	190	Cl—F	253
C—O	358	N—F	272	O—Cl	203	Cl—Cl	242
C—F	485	N—Cl	200	O—I	234		
C—Cl	328	N—Br	243			Br—F	237
C—Br	276			S—H	339	Br—Cl	218
C—I	240	H—H	436	S—F	327	Br—Br	193
C—S	259	H—F	567	S—Cl	253		
		H—Cl	431	S—Br	218	I—Cl	208
Si—H	323	H—Br	366	S—S	266	I—Br	175
Si—Si	226	H—I	299			I—I	151
Si—C	301						
Si—O	368						
Si—Cl	464						
Multiple Bonds							
C=C	614	N=N	418	O=O	495		
C≡C	839	N≡N	941				
C=N	615	N=O	607	S=O	523		
C≡N	891			S=S	418		
C=O	799						
C≡O	1072						

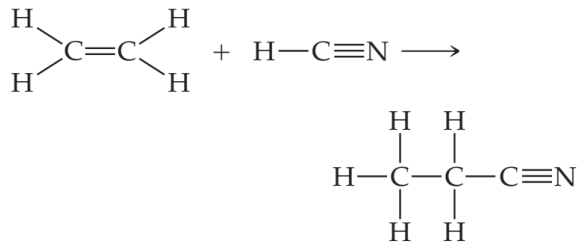
Exercise 7. *(Und)*

Using the above Table 8.4, estimate Δ*H* for each of the following gas- phase reactions (note that lone pairs on atoms are not shown) and comment on obtained results:

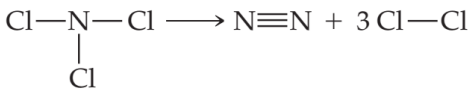
(a)



(b)

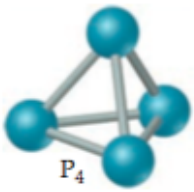


(c)



Exercise 8. ***(App)***

a) A common form of the elemental phosphorus is the tetrahedral P_4 molecule which is a solid at room temperature, where all four P atoms are equivalent:



a) Are there any lone paired of electrons in the P_4 molecule?

b) Draw a Lewis structure of a linear P_4 molecule that satisfies the octet rule. Does this molecule have resonance structures? On the basis of formal charges, which is more stable, the linear molecule or the tetrahedral molecule?

c) The reaction $P_4(g) \Leftrightarrow 2P_2(g)$ has ΔH = 217 KJ/mol. If the bond energy of a single P-P bond is 200 kJ/mol, calculate the bond energy of the P≡P bond.

d) Compare the value you obtain with the bond energy in N_2 (946 kJ/mol), and suggest an explanation for the difference in bond energies in P_2 and N_2.